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Table of Content

1 Introduction 3
2 Symbols and relationship between different notations 4
3 Units 5
4 Nomenclature 6
5 General PID structure 8
6 Transfer functions according to IEC 61362 nomenclature 11
6.1 Speed Control Mode 11
6.2 Power control mode 14
6.3 Power control mode with FCR 17
6.4 Power control mode FSM, LFSM 23
6.5 Flow control mode 29
6.6 Frequency influence PID loop (including FCR mode) 32
6.7 Control signal selection 34
7 Transfer functions according to IEEE 1207 nomenclature 36
7.1 Speed Control Mode 36
7.2 Power Control Mode 39
7.3 Flow Control Mode 42
7.4 Frequency Influence PID loop 45
7.5 Control signal selection 47

1 Introduction

This document provides transfer functions and models which are suitable to describe Voith HyCon Digital Turbine Governors as part of computer simulation models for the electric power system.
The transfer functions are represented according to the recommended practices defined by IEEE Guide for the Application of Turbine Governing Systems for Hydroelectric Generating Units, IEEE Std 1207™-2011 and IEC Guide to specification of hydraulic turbine governing systems, IEC 61362 Ed.2.0.
It is important to note that IEEE and IEC define only simplified general models and parameters for Hydro Units governing systems.

2 Symbols and relationship between different notations

International standards are using different symbols for elements of governors in hydropower plants. Below is a summary of the most important symbols and relationships between notations as defined in IEC 61362 Ed.2.0, IEC 60050-351, IEEE Std. 1207-2011 and ASME PTC 29-2005.

Symbols and correlation	**Description**
K_p	Proportional gain (IEEE 1207, ASME PTC 29) Proportional action coefficient (IEC 61362, IEC 60050-351) In German: Proportionalbeiwert (IEC 61362, IEC 60050-351)
b_p	Permanent speed droop (IEEE 1207, ASME PTC 29) Permanent droop (IEC 61362)
$K_i = K_I = \frac{K_p}{T_i}$	Integral gain K_i (IEEE 1207) Integral action coefficient K_I (IEC 61362, IEC 60050-351) In German: Integrierbeiwert K_I (IEC 61362, IEC 60050-351) Integral gain K_I (ASME PTC 29)
$T_i = \frac{K_p}{K_I} = K_p T_I = T_d$	Reset time T_i (IEC 61362, IEC 60050-351) In German: Nachstellzeit T_i (IEC 61362, IEC 60050-351) Damping time constant T_d (ASME PTC 29)
$T_I = \frac{1}{K_I}$	Integral action time (IEC 61362, IEC 60050-351) In German: Integrierzeit (IEC 61362, IEC 60050-351)
$K_d = K_D = K_{1D} T_{1D}$	Derivative gain K_d (IEEE 1207) Derivative action coefficient K_D (IEC 61362, IEC 60050-351) In German: Differenzierbeiwert K_D (IEC 61362, IEC 60050-351)
$T_D = K_D$	Derivative action time T_D (IEC 61362, IEC 60050-351) In German: Differenzierzeit T_D (IEC 61362, IEC 60050-351)
$K_D = K_{1D} \neq K_{D_{IEC61362}}$	Derivative gain K_D (ASME PTC 29)
$T_d = \frac{K_D}{K_p} = \frac{T_D}{K_p}$	Rate time T_d (IEC 61362, IEC 60050-351) In German: Vorhaltzeit T_d (IEC 61362, IEC 60050-351)
$T_d = T_{1D}$	Time constant of the damping device T_d (IEEE 1207)
$T_V = T_d = T_{1D}$	Derivative Filter Time Constant T_V, T_d (ASME PTC 29)

3 Units

The parameters and variables of the models (with the exception of head) are represented in percent with reference values according to the following table:

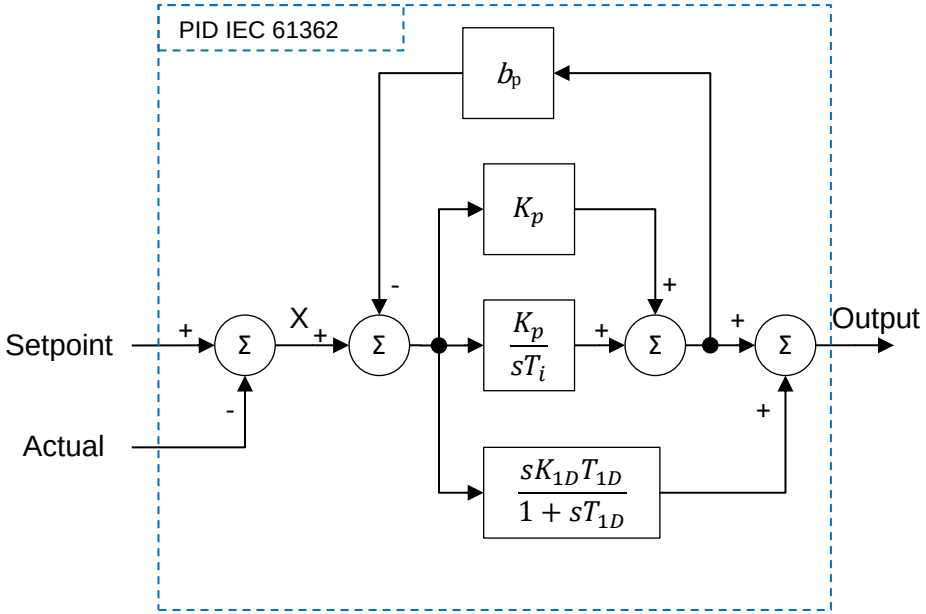
Values	**Unit**	**Percent**	**Reference value**
Active Power	MW	% of rated power	Turbine rated power
Main Distributing Valve Position	-	%	Maximum amplitude
Speed	rpm	% of rated speed	Shaft rated speed
Grid Frequency	Hz	% of nominal frequency	Grid nominal frequency
Turbine Head	m	Not applicable	Turbine rated head
Water level	m	No general definition available	Sometimes used: max – min level
Units and reference values			

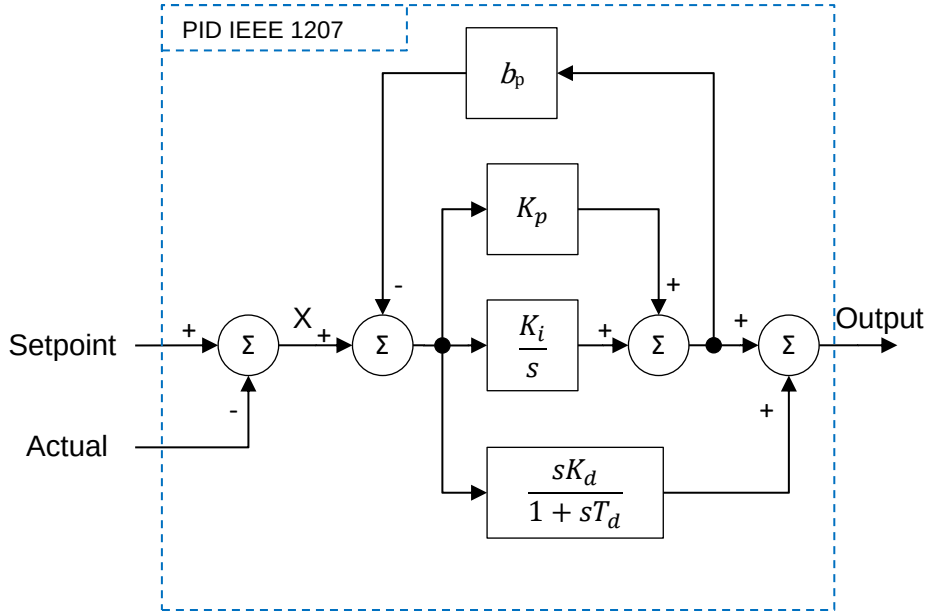
4 Nomenclature

ACCL	Acceleration control
AFC	Frequency influence (automatic frequency control)
CAVN	Cavitation
DEFL	Deflector
$DEFL_{ONL}$	Deflector at Speed No Load (%)
$DEFL_{sp}$	Deflector Control Setpoint (%)
F_{nom}	Nominal Grid Frequency (%)
F_{act}	Actual Grid Frequency (%)
$F_{actFiltered}$	Actual Filtered Grid Frequency (%)
FCR	Frequency Containment Reserve (including FCR-N and FCR-D)
FCR-N	Frequency Containment Reserve for Normal Operation
FCR-D	Frequency Containment Reserve for Disturbances
FCR-D up	Frequency Containment Reserve for Disturbances upwards regulation ($F_{act} < F_{nom}$)
FCR-D down	Frequency Containment Reserve for Disturbances downwards regulation ($F_{act} > F_{nom}$)
FLOW	Flow control
$FLOW_{act}$	Actual Flow(%)
$FLOW_{ONL}$	Flow at Speed No Load (%)
$FLOW_{sp}$	Flow Control Setpoint (%)
$FLOW\text{-}LMT_{sp}$	Flow Limitation Setpoint (%)
FSML	Frequency Sensitive Mode and Limited Frequency Sensitive Mode
FSM	Frequency Sensitive Mode
LFSM	Limited Frequency Sensitive Mode
LFSM-U	Limited Frequency Sensitive Mode – Underfrequency ($F_{act} < F_{nom}$)
LFSM-O	Limited Frequency Sensitive Mode – Overfrequency ($F_{act} > F_{nom}$)
MV	Actual Main Distributing Valve position (%)
NOZ	Nozzle
OVRLD	Overload
POSN	Position control
POW	Power control
POW_{sp}	Power Control Setpoint (%)

POW_{act}	Actual Power (%)
POW-FF	Power control feed-forward curve
$POW\text{-}FLOW_{sp}$	Flow Setpoint from Power Control (Power Control Loop Output) (%)
POW-LMT	Power limit control
$POW\text{-}LMT_{sp}$	Power Limitation Setpoint (%)
PP	Pump operation
RB	Runner blade
SC	Synchronous condenser
SPD	Speed control
SPD_{sp}	Speed Control Setpoint (%)
SPD_{act}	Actual Speed (%)
SPD-NO-LD	Speed control no-load
SPD-ISOL	Speed control isolated grid
SPD-GD	Speed control large grid
SPD-BS	Speed control blackstart
$SPD\text{-}DEFL_{sp}$	Deflector Setpoint from Speed Control (Speed Control Loop Output) (%)
$SPD\text{-}FLOW_{sp}$	Flow Setpoint from Speed Control (Speed Control Loop Output) (%)
SRG	Surge control
WG	Wicket gate
HyCon Governor Nomenclature	

5 General PID structure





Note: PID structure is different concerning permanent speed droop and derivative part compared to example structure of figure 9 of IEEE 1207-2011

Parameters		Comments
bp	Permanent droop	IEC 61362, IEEE1207
K_p	Proportional gain	IEC 61362, IEEE1207
T_i	Reset time	IEC 61362
K_{1D}	Derivative gain	IEC 61362
T_{1D}	Derivative time constant	IEC 61362
K_i	Integral gain	IEEE1207
K_d	Derivative gain	IEEE1207
T_d	Time constant of the damping device	IEEE1207
General PID parameters		

Conversion rules:

$b_{p\,\mathrm{IEEE1207}} = b_{p\,\mathrm{IEC61362}}$

$K_{p\,\mathrm{IEEE1207}} = K_{p\,\mathrm{IEC61362}}$

$K_{i\,\mathrm{IEEE1207}} = K_{p\,\mathrm{IEC61362}} / T_{i\,\mathrm{IEC61362}}$

$K_{d\,\mathrm{IEEE1207}} = K_{1D\,\mathrm{IEC61362}} * T_{1D\,\mathrm{IEC61362}}$

$T_{d\,\mathrm{IEEE1207}} = T_{1D\,\mathrm{IEC61362}}$

Transfer function PID according to IEC 61362 nomenclature

$$\frac{Output}{X} = \frac{K_p + \frac{K_p}{s * T_i} + \frac{s * K_{1D} * T_{1D}}{1 + s * T_{1D}}}{1 + b_p \left(K_p + \frac{K_p}{s * T_i}\right)}$$

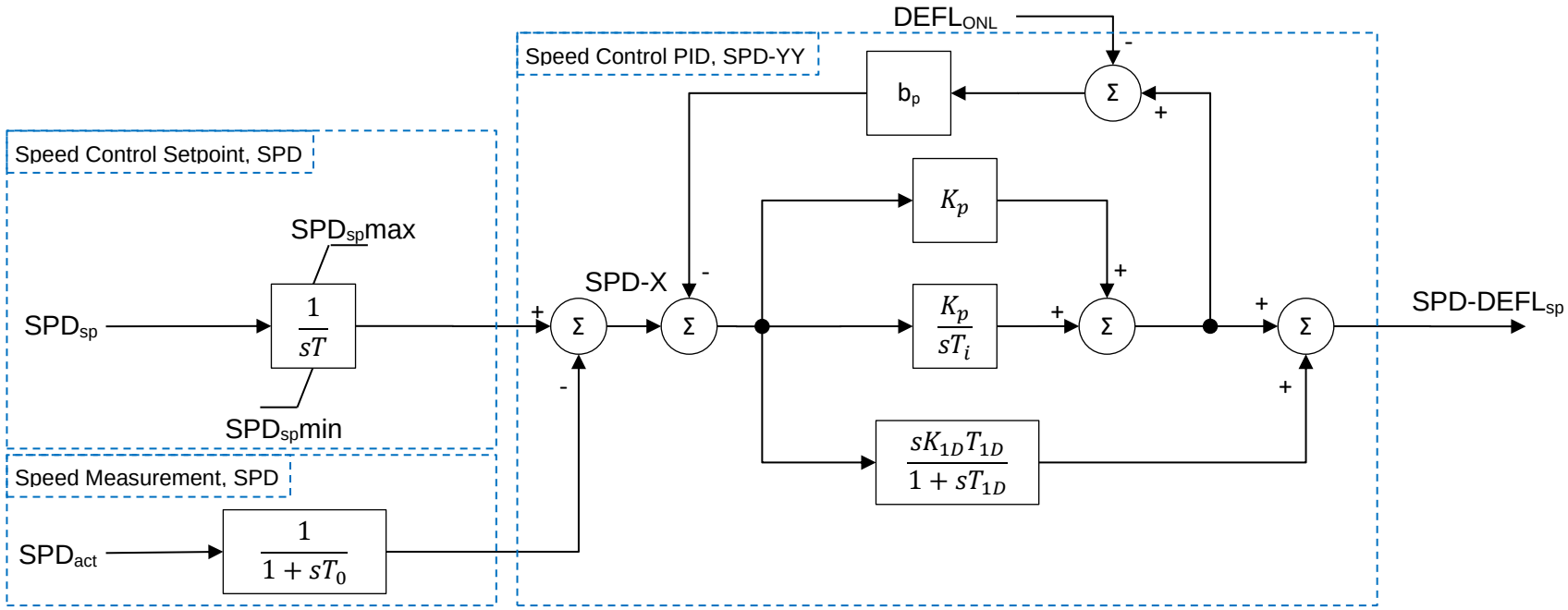
Transfer function PID according to IEEE 1207 nomenclature

$$\frac{Output}{X} = \frac{K_p + \frac{K_i}{s} + \frac{s * K_d}{1 + s * T_d}}{1 + b_p \left(K_p + \frac{K_i}{s}\right)}$$

Note: PID structure is different concerning permanent speed droop and derivative part compared to example structure of figure 9 of IEEE 1207-2011.

6 Transfer functions according to IEC 61362 nomenclature

6.1 Speed Control Mode

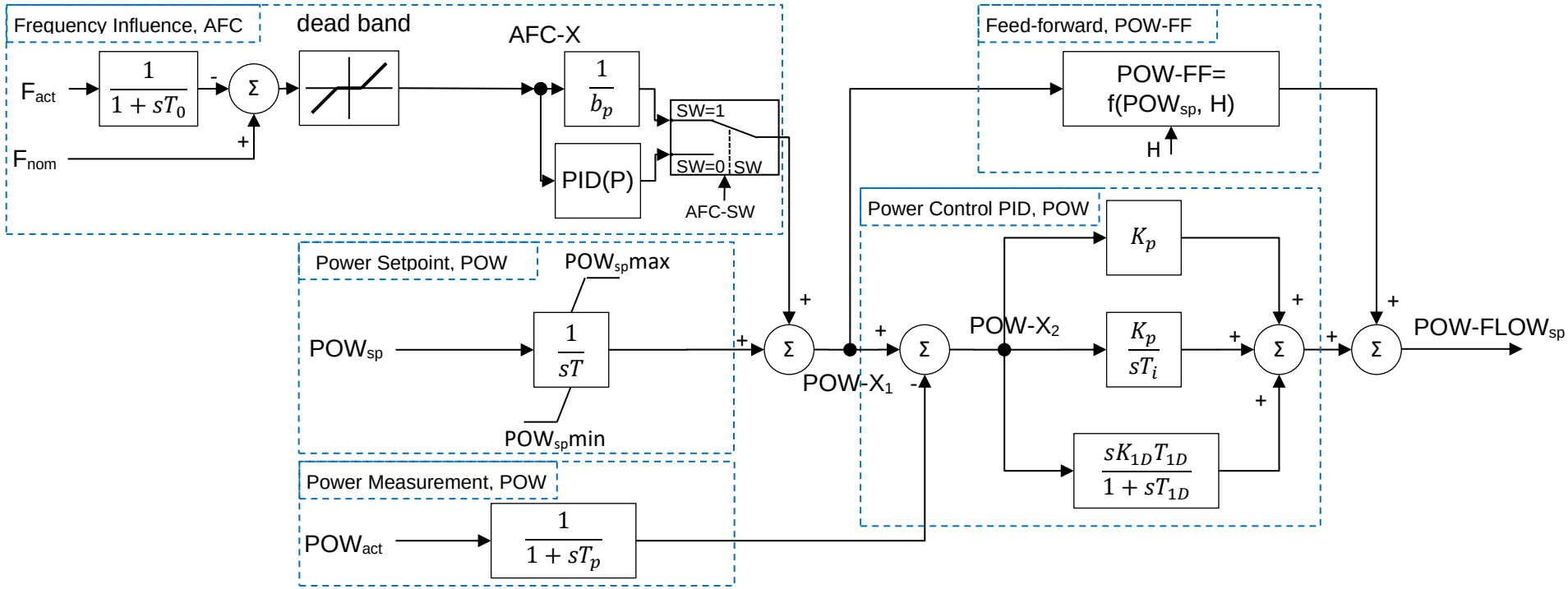


Parameters		Parameter range	Typical	Comments
$SPD_{sp}min$	Minimum allowed Speed set point	0% - 200%	90%	
$SPD_{sp}max$	Maximum allowed Speed set point	0% - 200%	110%	
SPD-T	Setpoint ramp time 0%-100% for Speed Control mode [s]	10 -120s	60s	
$SPD\text{-}T_0$	Speed Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$SPD\text{-}YY\text{-}b_p$	Permanent droop	0 – 0.1 (0% - 10%)	0 (0%) – No Load (YY= NO-LD) 0.03 (3%) – Isolated (YY= ISOL) 0.05 (5%) – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}K_p$	Proportional gain	0.1 – 20	1.2 – No Load (*YY*= NO-LD) 0.9 – Isolated (*YY*= ISOL) 0.8 – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}T_i$	Reset time [s]	0.1 – 100s	6.0s – No Load (*YY*= NO-LD) 7.0s – Isolated (*YY*= ISOL) 5.0s – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}K_{1D}$	Derivative gain	0 – 10	0.0 – No Load (*YY*= NO-LD) 0.0 – Isolated (*YY*= ISOL) 0.0 – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}T_{1D}$	Derivative time constant [s]	0.1 – 10s	2s (*YY*= NO-LD, ISOL, BS)	
Speed control mode parameters				

Transfer function Speed Control PID ($DEFL_{ONL}$ = constant):

$$\frac{SPD\text{-}DEFL_{sp}}{SPD\text{-}X} = \frac{SPD\text{-}K_p + \frac{SPD\text{-}K_p}{s * SPD\text{-}T_i} + \frac{s * SPD\text{-}K_{1D} * SPD\text{-}T_{1D}}{1 + s * SPD\text{-}T_{1D}}}{1 + SPD\text{-}b_p \left(SPD\text{-}K_p + \frac{SPD\text{-}K_p}{s * SPD\text{-}T_i} \right)}$$

6.2 Power control mode



Parameters		Parameter Range	Typical	Comments
$POW_{sp}min$	Minimum allowed Power set point	0 – 120%	0%	
$POW_{sp}max$	Maximum allowed Power set point	0 – 120%	100%	
$AFC\text{-}T_0$	Frequency Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$AFC\text{-}b_p$	Frequency influence (droop) bp	0.001 – 0.1 (0.1% - 10%)	0.05 (5%)	
$POW\text{-}T_p$	Power Measurement Filter Time Constant [s]	0.02 – 1s	0.2s	
$POW\text{-}T$	Setpoint ramp time 0%-100% for Power Control mode [s]	10 – 120s	60s	
$POW\text{-}K_p$	Proportional gain	0.1 – 20	0.1	
$POW\text{-}T_i$	Reset time [s]	0.1 – 100s	6s	
$POW\text{-}K_{1D}$	Derivative gain	0 – 10	0	
$POW\text{-}T_{1D}$	Derivative time constant [s]	0.1 – 10s	1s	
POW-FF	Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point)			
AFC-SW	Frequency influence software switch	0 - 1	1	0 = *PID* 1 = bp
Power control mode parameters				

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

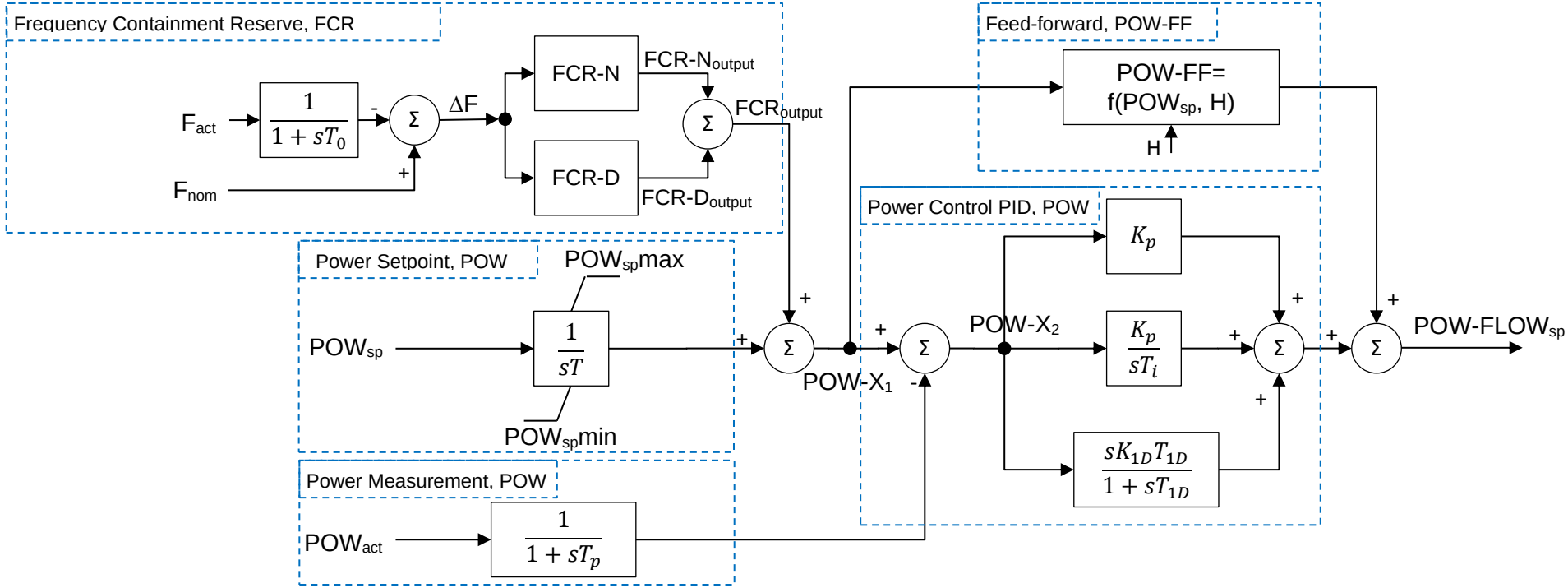
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

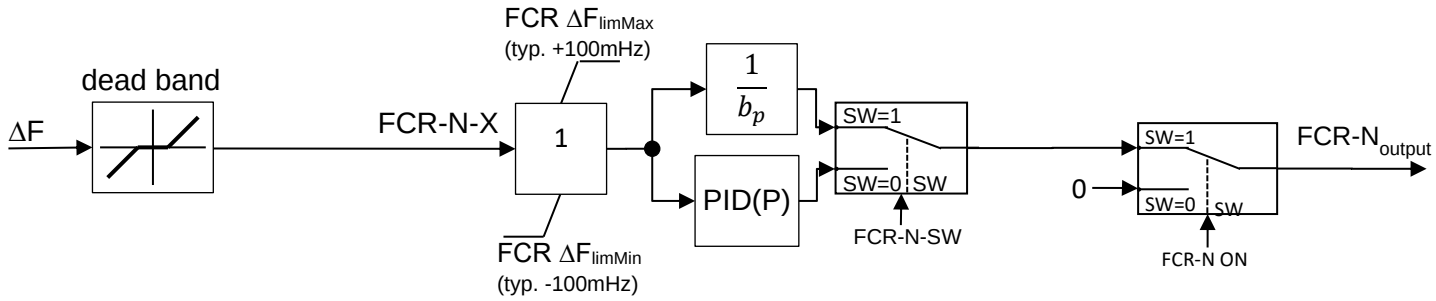
Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} + \frac{s * AFC\text{-}K_{1D} * AFC\text{-}T_{1D}}{1 + s * AFC\text{-}T_{1D}}}{1 + AFC\text{-}b_p * \left(AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

6.3 Power control mode with FCR

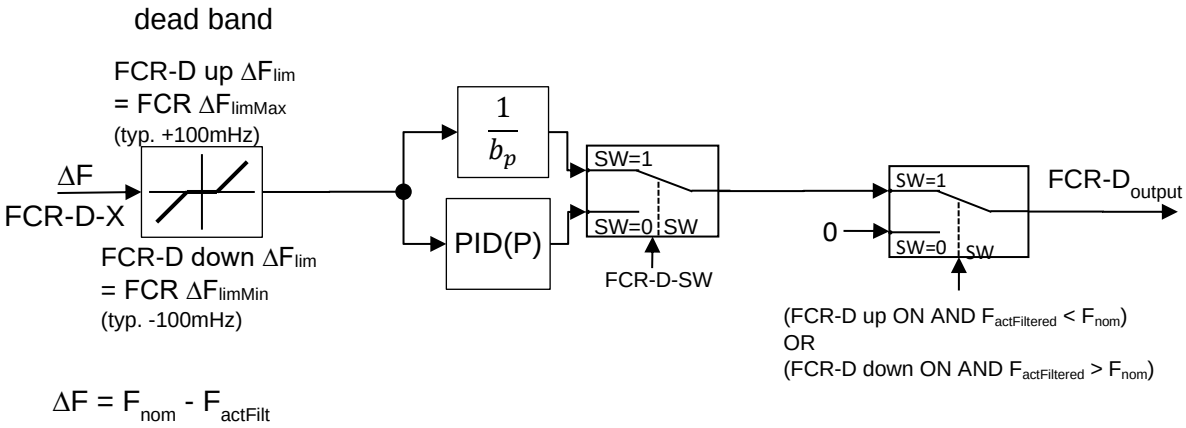


Frequency Containment Reserve Normal Operation, FCR-N



$\Delta F = F_{nom} - F_{actFilt}$

Frequency Containment Reserve for Disturbances, FCR-D



Parameters		Parameter Range	Typical	Comments
$POW_{sp}min$	Minimum allowed Power set point	0 – 120%	0%	
$POW_{sp}max$	Maximum allowed Power set point	0 – 120%	100%	
$FCR\text{-}T_0$	Frequency Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$FCR\text{-}N\text{-}b_p$	FCR-N (droop) bp	0.001 – 0.1 (0.1% - 10%)	0.05 (5%)	
$FCR\text{-}D\text{-}b_p$	FCR-D (droop) bp	0.001 – 0.1 (0.1% - 10%)	0.05 (5%)	
$POW\text{-}T_p$	Power Measurement Filter Time Constant [s]	0.02 – 1s	0.2s	
$POW\text{-}T$	Setpoint ramp time 0%-100% for Power Control mode [s]	10 – 120s	60s	
$POW\text{-}K_p$	Proportional gain	0.1 – 20	0.1	
$POW\text{-}T_i$	Reset time [s]	0.1 – 100s	6s	
$POW\text{-}K_{1D}$	Derivative gain	0 – 10	0	
$POW\text{-}T_{1D}$	Derivative time constant [s]	0.1 – 10s	1s	
POW-FF	Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point)			
FCR-N-SW	FCR-N software switch PID/droop	0 - 1	0	0 = *PID* 1 = bp
FCR-D-SW	FCR-D software switch PID/droop	0 - 1	0	0 = *PID* 1 = bp
Power control mode with FCR-N and FCR-D parameters				

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-N with static droop ($|\Delta F| \leq |FCR\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}N\text{-}X} = \frac{1}{FCR\text{-}N\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-N with PID ($|\Delta F| \leq |FCR\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}N\text{-}X} = \frac{FCR\text{-}N\text{-}K_p + \frac{FCR\text{-}N\text{-}K_p}{s * FCR\text{-}N\text{-}T_i} + \frac{s * FCR\text{-}N\text{-}K_{1D} * FCR\text{-}N\text{-}T_{1D}}{1 + s * FCR\text{-}N\text{-}T_{1D}}}{1 + FCR\text{-}N\text{-}b_p * \left(FCR\text{-}N\text{-}K_p + \frac{FCR\text{-}N\text{-}K_p}{s * FCR\text{-}N\text{-}T_i}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-D with static droop ($|\Delta F| > |FCR\ \Delta F_{lim}|$)

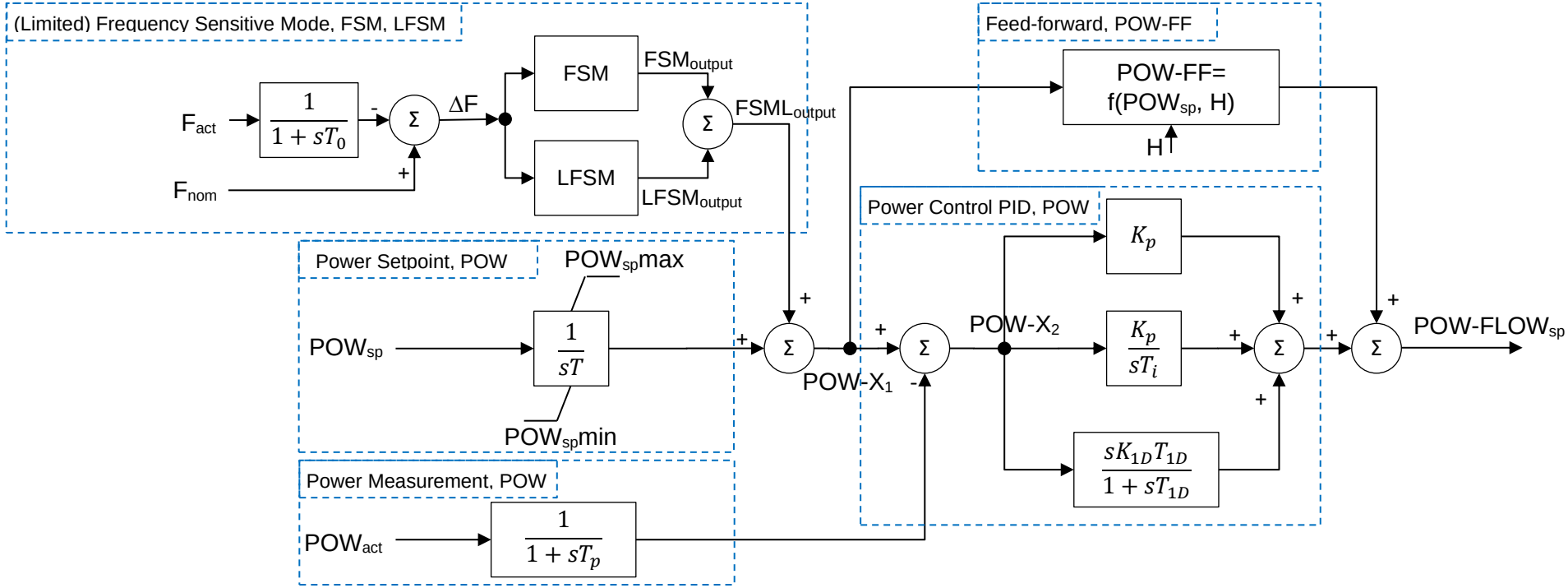
$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}D\text{-}X} = \left[\frac{1}{FCR\text{-}N\text{-}b_p} * \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X} + \frac{1}{FCR\text{-}D\text{-}b_p} * \left(1 - \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X}\right)\right] * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FCR-D with PID ($|\Delta F| > |FCR\ \Delta F_{lim}|$)

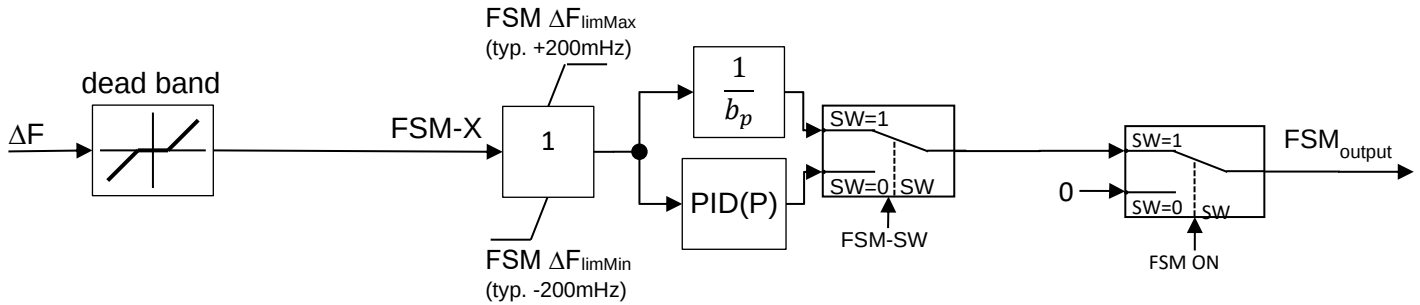
$$\frac{POW\text{-}FLOW_{sp}}{FCR\text{-}D\text{-}X} = \left[\frac{1}{FCR\text{-}N\text{-}b_p} * \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X} + \frac{FCR\text{-}D\text{-}K_p + \frac{FCR\text{-}D\text{-}K_p}{s * FCR\text{-D-}T_i} + \frac{s * FCR\text{-}D\text{-}K_{1D} * FCR\text{-}D\text{-}T_{1D}}{1 + s * FCR\text{-}D\text{-}T_{1D}}}{1 + FCR\text{-}D\text{-}b_p * \left(FCR\text{-}D\text{-}K_p + \frac{FCR\text{-}D\text{-}K_p}{s * FCR\text{-}D\text{-}T_i}\right)} * \left(1 - \frac{FCR\Delta F_{lim}}{FCR\text{-}D\text{-}X}\right)\right]$$

$$* \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

6.4 Power control mode FSM, LFSM

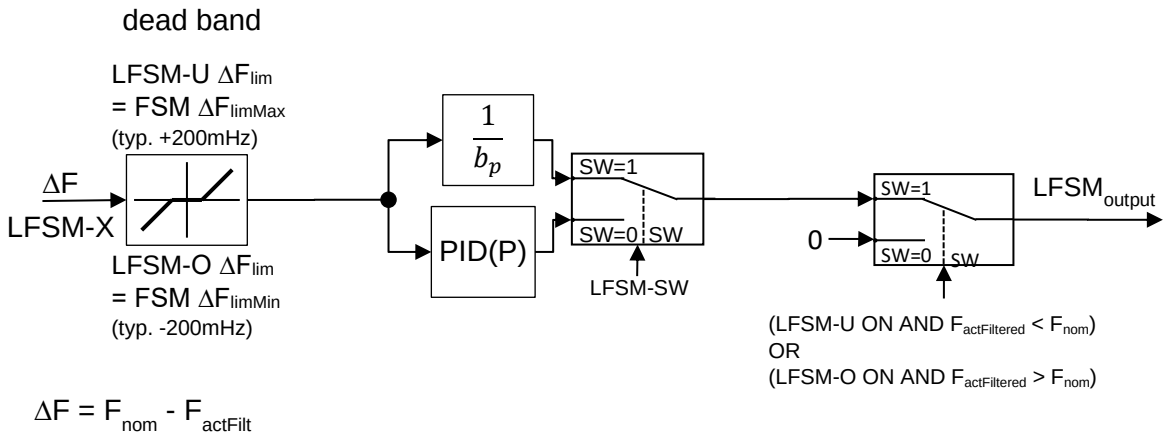


Frequency Sensitive Mode, FSM



$\Delta F = F_{nom} - F_{actFilt}$

Limited Frequency Sensitive Mode, LFSM



Parameters		Parameter Range	Typical	Comments
$POW_{sp}min$	Minimum allowed Power set point	0 – 120%	0%	
$POW_{sp}max$	Maximum allowed Power set point	0 – 120%	100%	
$FSM\text{-}T_0$	Frequency Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$FSM\text{-}b_p$	FSM (droop) bp	0.001 – 0.1 (0.1% - 10%)	0.05 (5%)	
$LFSM\text{-}b_p$	LFSM (droop) bp	0.001 – 0.1 (0.1% - 10%)	0.05 (5%)	
$POW\text{-}T_p$	Power Measurement Filter Time Constant [s]	0.02 – 1s	0.2s	
$POW\text{-}T$	Setpoint ramp time 0%-100% for Power Control mode [s]	10 – 120s	60s	
$POW\text{-}K_p$	Proportional gain	0.1 – 20	0.1	
$POW\text{-}T_i$	Reset time [s]	0.1 – 100s	6s	
$POW\text{-}K_{1D}$	Derivative gain	0 – 10	0	
$POW\text{-}T_{1D}$	Derivative time constant [s]	0.1 – 10s	1s	
POW-FF	Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point)			
FSM-SW	FSM software switch PID/droop	0 - 1	0	0 = *PID* 1 = bp
LFSM-SW	LFSM software switch PID/droop	0 - 1	0	0 = *PID* 1 = bp
Power control mode with FSM/LFSM parameters				

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FSM with static droop ($|\Delta F| \leq |FSM\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FSM\text{-}X} = \frac{1}{FSM\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for FSM with PID ($|\Delta F| \leq |FSM\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{FSM\text{-}X} = \frac{FSM\text{-}K_p + \frac{FSM\text{-}K_p}{s * FSM\text{-}T_i} + \frac{s * FSM\text{-}K_{1D} * FSM\text{-}T_{1D}}{1 + s * FSM\text{-}T_{1D}}}{1 + FSM\text{-}b_p * \left(FSM\text{-}K_p + \frac{FSM\text{-}K_p}{s * FSM\text{-}T_i}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for LFSM with static droop ($|\Delta F| > |FSM\ \Delta F_{lim}|$)

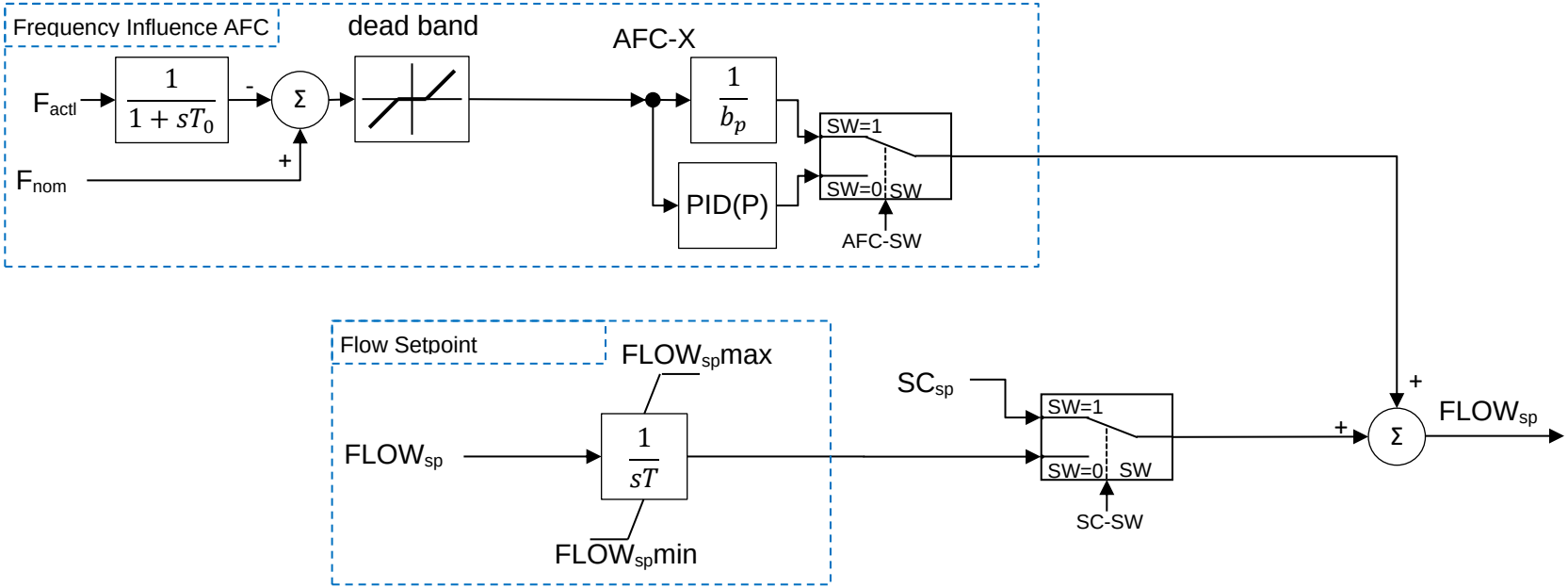
$$\frac{POW\text{-}FLOW_{sp}}{LFSM\text{-}X} = \left[\frac{1}{FSM\text{-}b_p} * \frac{FSM\Delta F_{lim}}{LFSM\text{-}X} + \frac{1}{LFSM\text{-}b_p} * \left(1 - \frac{FSM\Delta F_{lim}}{LFSM\text{-}X}\right)\right] * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

Transfer function for evaluation of the response to frequency disturbances for LFSM with PID ($|\Delta F| > |FSM\ \Delta F_{lim}|$)

$$\frac{POW\text{-}FLOW_{sp}}{LFSM\text{-}X} = \left[\frac{1}{FSM\text{-}b_p} * \frac{FSM\Delta F_{lim}}{LFSM\text{-}X} + \frac{LFSM\text{-}K_p + \frac{LFSM\text{-}K_p}{s * LFSM\text{-}T_i} + \frac{s * LFSM\text{-}K_{1D} * LFSM\text{-}T_{1D}}{1 + s * LFSM\text{-}T_{1D}}}{1 + LFSM\text{-}b_p * \left(LFSM\text{-}K_p + \frac{LFSM\text{-}K_p}{s * LFSM\text{-}T_i}\right)} * \left(1 - \frac{FSM\Delta F_{lim}}{LFSM\text{-}X}\right)\right]$$

$$* \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_p}{s * POW\text{-}T_i} + \frac{s * POW\text{-}K_{1D} * POW\text{-}T_{1D}}{1 + s * POW\text{-}T_{1D}}\right)$$

6.5 Flow control mode



Parameters		Parameter Range	Typical	Comments
$FLOW_pmin$	Minimum allowed Flow set point	0 – 100%	0%	
$FLOW_{sp}max$	Maximum allowed Flow set point	0 – 100%	100%	
FLOW-T	Setpoint ramp time 0%-100% for Flow Control mode [s]	10 – 120s	60s	
AFC-T_0	Speed Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
AFC-b_p	Frequency influence (droop) bp	0.001 – 0.1 (0.1% – 10%)	0.05 (5%)	
AFC-SW	Frequency influence software switch	0 – 1	1	0 = PID 1 = bp
SC-SW	Synchronous condenser software switch	0 – 1	0	0 = Off 1 = On
SC_{sp}	Flow set point in Synchronous condenser mode	0 – 100%	0%	
Flow control mode parameters				

Note: AFC can also stand for FCR (see chapter *Power control mode with FCR*) or FSML (see chapter *Power control mode with FSM*)

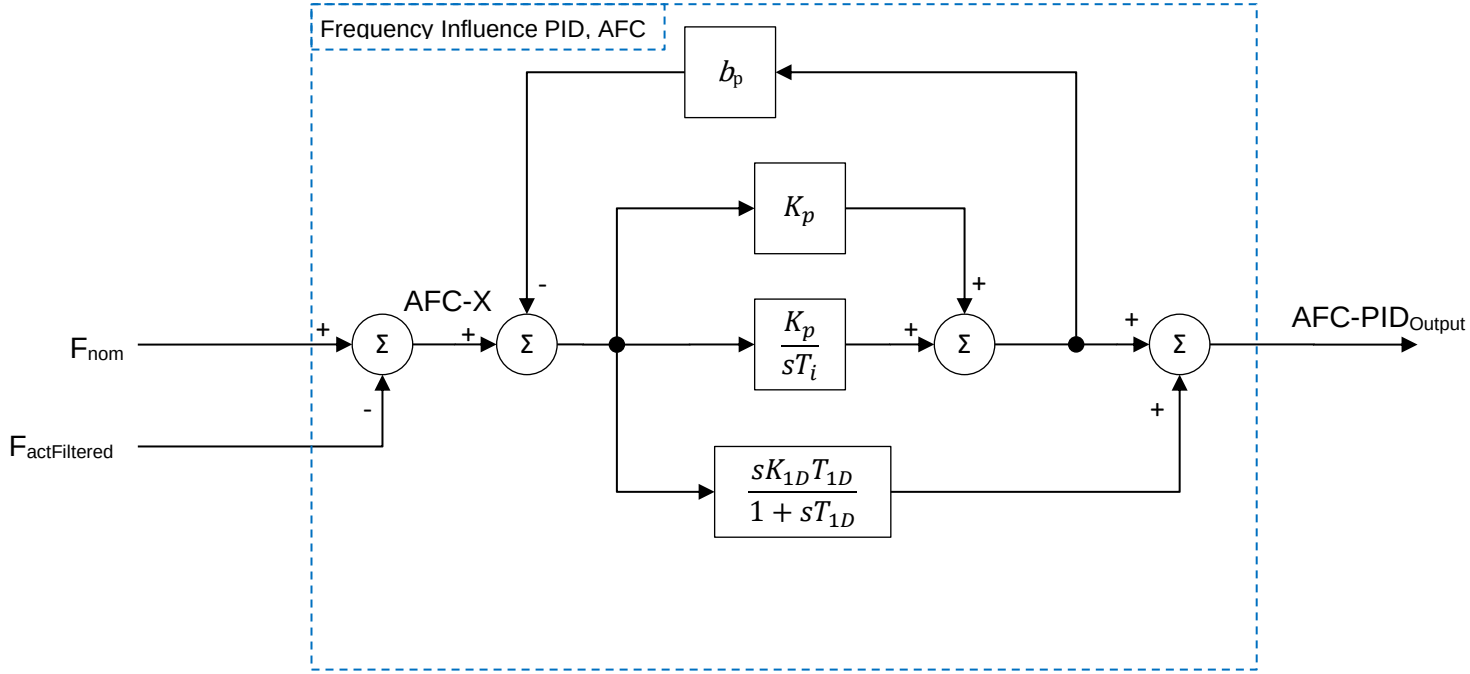
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p}$$

Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} + \frac{s * AFC\text{-}K_{1D} * AFC\text{-}T_{1D}}{1 + s * AFC\text{-}T_{1D}}}{1 + AFC\text{-}b_p \left(AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} \right)}$$

6.6 Frequency influence PID loop (including FCR mode)



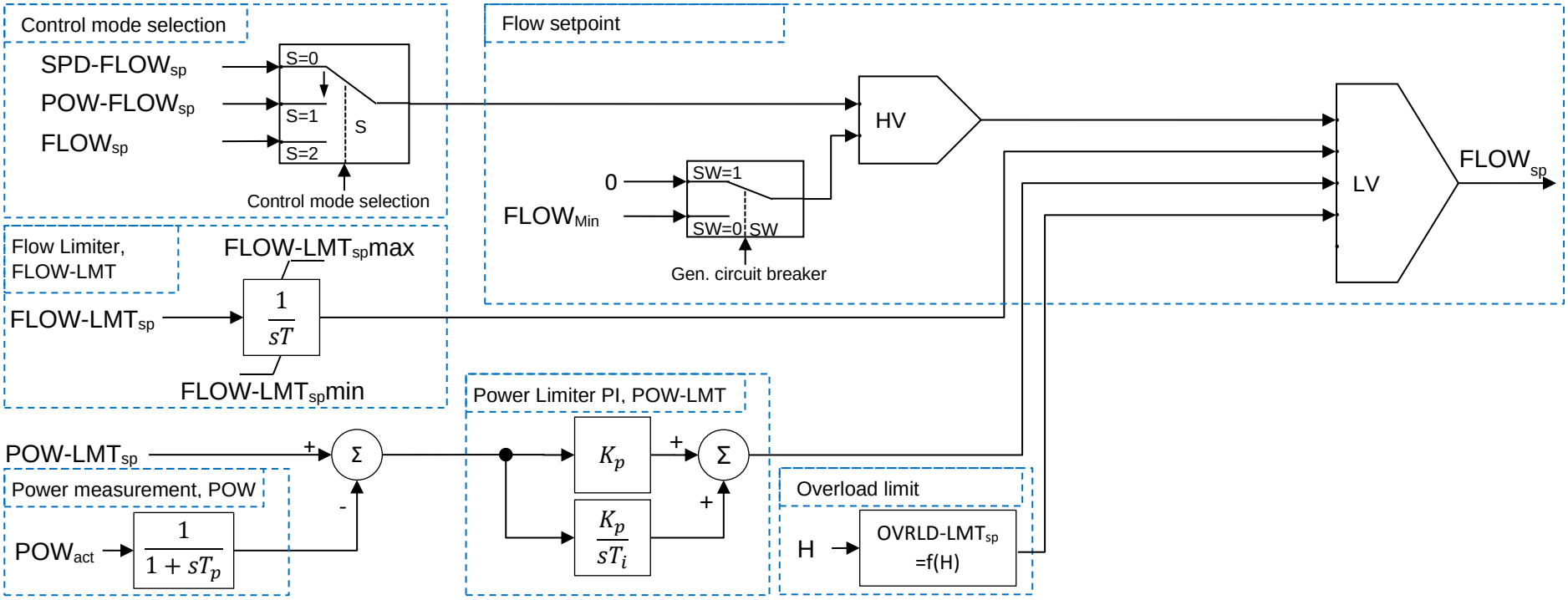
Parameters		Parameter Range	Typical	Comments
$AFC\text{-}bp$	Permanent droop	0 – 0.1 (0% - 10%)	0.05 (5%)	
$AFC\text{-}K_p$	Proportional gain	0.1 – 20	1.5	
$AFC\text{-}T_i$	Reset time [s]	0.1 – 100s	3s	
$AFC\text{-}K_{1D}$	Derivative gain	0 – 10	0	
$AFC\text{-}T_{1D}$	Derivative time constant [s]	0.1 – 10s	0s	
Frequency influence PID parameters				

Transfer function Frequency Influence PID:

$$\frac{AFC\text{-}PID_{Output}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i} + \frac{s * AFC\text{-}K_{1D} * AFC\text{-}T_{1D}}{1 + s * AFC\text{-}T_{1D}}}{1 + AFC\text{-}b_p\left(AFC\text{-}K_p + \frac{AFC\text{-}K_p}{s * AFC\text{-}T_i}\right)}$$

Note: AFC can also stand for FCR-N and FCR-D (see chapter *Power control mode with FCR*) or FSM and LFSM (see chapter *Power control mode with FSM, LFSM*)

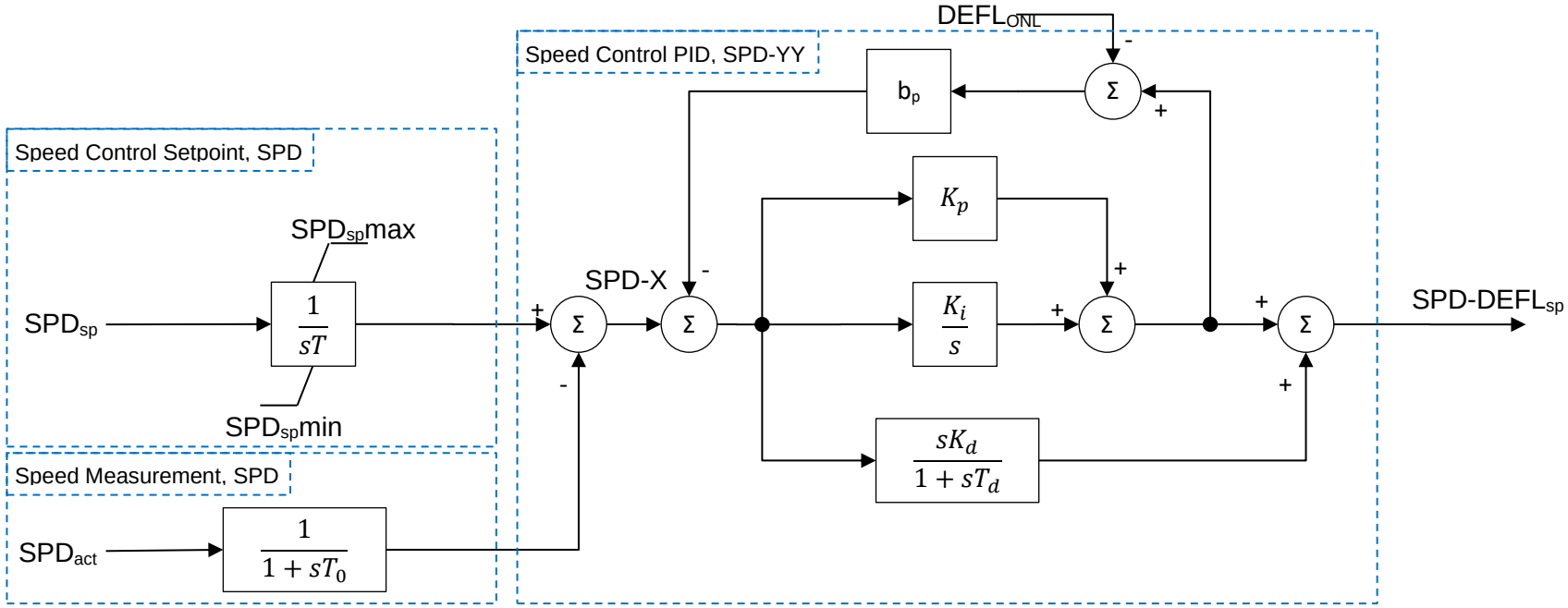
6.7 Control signal selection



Parameters		Parameter Range	Typical	Comments
FLOW-LMT$_{sp}$min	Minimum allowed Flow Limiter set point	0 - 100%	0%	
FLOW-LMT$_{sp}$max	Maximum allowed Flow Limiter set point	0 - 100%	100%	
FLOW-LMT-T	Setpoint ramp time 0%-100% for Flow Limiter [s]	10 – 120s	60s	
POW-T_p	Power Measurement Filter Time Constant [s]	0.02 – 1s	0.2s	
FLOW$_{Min}$	Minimum flow with generator circuit breaker closed	0 – 10%	2% + O_{NL}	
POW-LMT-K_p	Proportional gain – Power Limiter	0.1 – 20	0.8	
POW-LMT-T_i	Reset time – Power Limiter [s]	0.1 – 100s	6s	
POW-LMT$_{sp}$	Maximum Power Limit	0 - 100%	100%	
OVRLD-LMT$_{sp}$	Corresponding overload flow limit to Head H	-		11 interpolated points
H	Actual head	-		
LV	Low value (Lowest value is transferred to output)	-		
HV	High value (Highest value is transferred to output)	-		
Control signal selection parameters				

7 Transfer functions according to IEEE 1207 nomenclature

7.1 Speed Control Mode

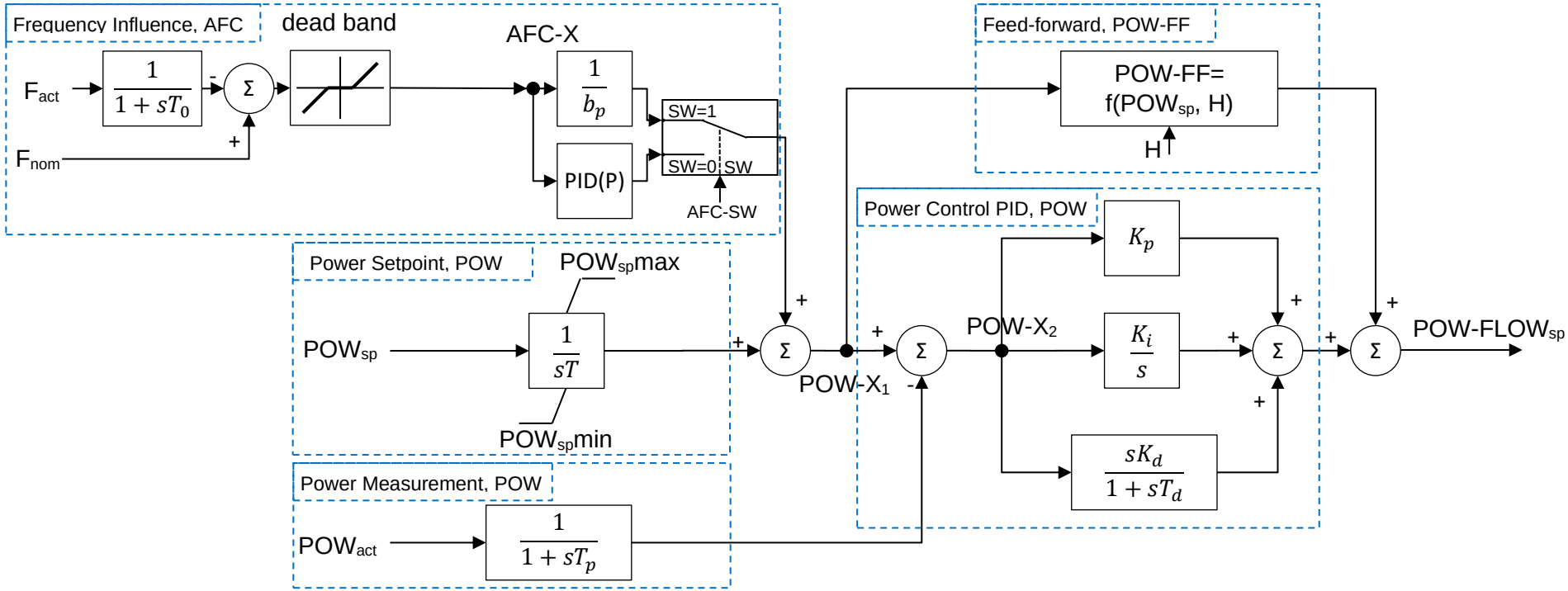


Parameters		Parameter range	Typical	Comments
$SPD_{sp}min$	Minimum allowed Speed set point	0% - 200%	90%	
$SPD_{sp}max$	Maximum allowed Speed set point	0% - 200%	110%	
SPD-T	Setpoint ramp time 0%-100% for Speed Control mode [s]	10 -120s	60s	
$SPD\text{-}T_0$	Speed Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$SPD\text{-}YY\text{-}b_p$	Permanent droop	0 – 0.1 (0% - 10%)	0 (0%) – No Load (YY= NO-LD) 0.03 (3%) – Isolated (YY= ISOL) 0.05 (5%) – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}K_p$	Proportional gain	0.1 – 20	1.2 – No Load (*YY*= NO-LD) 0.9 – Isolated (*YY*= ISOL) 0.8 – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}K_i$	Integral gain	0.01 – 100	0.2 – No Load (*YY*= NO-LD) 0.13 – Isolated (*YY*= ISOL) 0.16 – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}K_d$	Derivative gain	0 – 10	0.0 – No Load (*YY*= NO-LD) 0.0 – Isolated (*YY*= ISOL) 0.0 – Blackstart (*YY*= BS)	
$SPD\text{-}YY\text{-}T_d$	Time constant of the damping device [s]	0.1 – 10s	2s (*YY*= NO-LD, ISOL, BS)	
Speed control mode parameters				

Transfer function Speed Control PID ($DEFL_{ONL}$ = constant):

$$\frac{SPD\text{-}DEFL_{sp}}{SPD\text{-}X} = \frac{SPD\text{-}K_p + \frac{SPD\text{-}K_i}{s} + \frac{s * SPD\text{-}K_d}{1 + s * SPD\text{-}T_d}}{1 + SPD\text{-}b_p \left(SPD\text{-}K_p + \frac{SPD\text{-}K_i}{s}\right)}$$

7.2 Power Control Mode



Parameters		Parameter Range	Typical	Comments
$POW_{sp}min$	Minimum allowed Power set point	0 – 120%	0%	
$POW_{sp}max$	Maximum allowed Power set point	0 – 120%	100%	
$AFC\text{-}T_0$	Frequency Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$AFC\text{-}b_p$	Frequency influence (droop) bp	0.001 – 0.1 (0.1% - 10%)	0.05 (5%)	
$POW\text{-}T_p$	Power Measurement Filter Time Constant [s]	0.02 – 1s	0.2s	
$POW\text{-}T$	Setpoint ramp time 0%-100% for Power Control mode [s]	10 – 120s	60s	
$POW\text{-}K_p$	Proportional gain	0.1 – 20	0.1	
$POW\text{-}K_i$	Integral gain	0.01 – 100	0.02	
$POW\text{-}K_d$	Derivative gain	0 – 10	0	
$POW\text{-}T_d$	Time constant of the damping device [s]	0.1 – 10s	1s	
POW-FF	Corresponding opening to Head and Power setpoint (=dy/dP for actual operating point)			
AFC-SW	Frequency influence software switch	0 - 1	1	0 = *PID* 1 = bp
Power control mode parameters				

Transfer function for evaluation of the response to power disturbances

$$\frac{POW\text{-}FLOW_{sp}}{POW\text{-}X_2} = POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}$$

$$\frac{POW\text{-}FLOW_{sp}}{POW_{act}} = \frac{-1}{1 + s * POW\text{-}T_p} * \left(POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}\right)$$

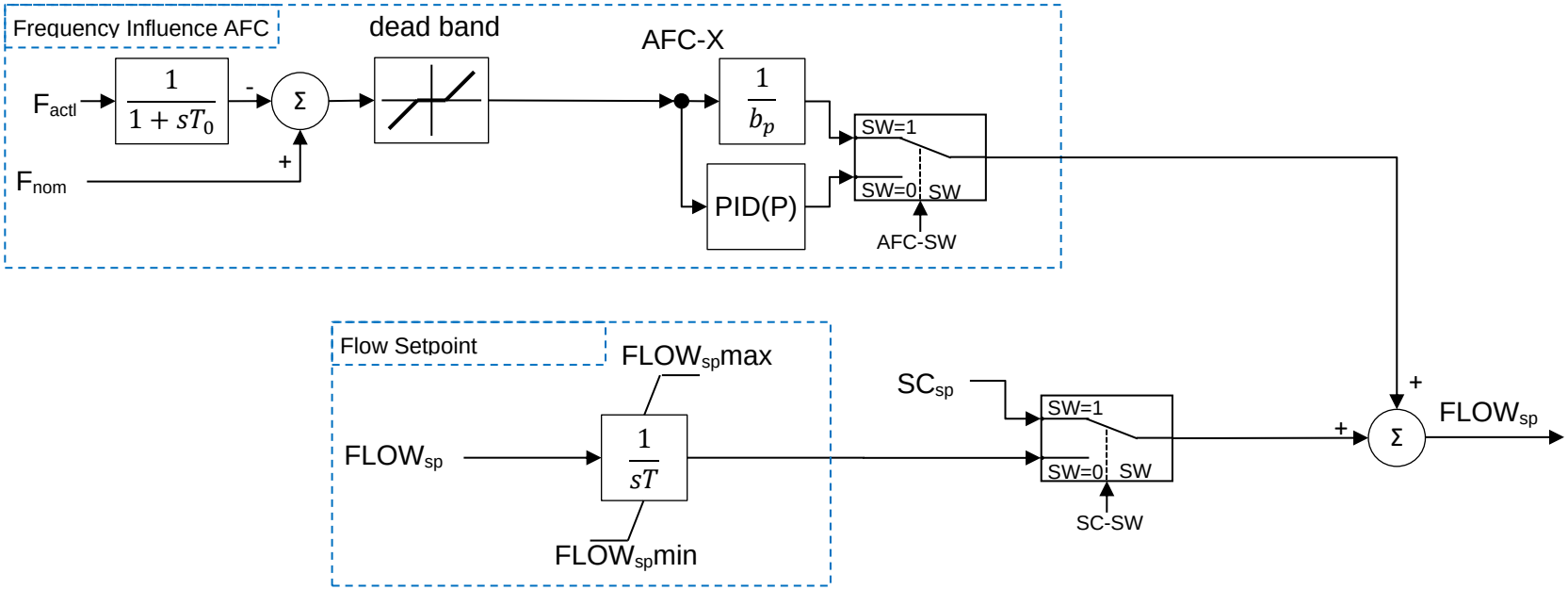
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}\right)$$

Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{POW\text{-}FLOW_{sp}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s} + \frac{s * AFC\text{-}K_d}{1 + s * AFC\text{-}T_d}}{1 + AFC\text{-}b_p * \left(AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s}\right)} * \left(\frac{\mathrm{dy}}{\mathrm{dP}} + POW\text{-}K_p + \frac{POW\text{-}K_i}{s} + \frac{s * POW\text{-}K_d}{1 + s * POW\text{-}T_d}\right)$$

7.3 Flow Control Mode



Parameters		Parameter Range	Typical	Comments
$FLOW_{sp}min$	Minimum allowed Flow set point	0 – 100%	0%	
$FLOW_{sp}max$	Maximum allowed Flow set point	0 – 100%	100%	
FLOW-T	Setpoint ramp time 0%-100% for Flow Control mode [s]	10 – 120s	60s	
$AFC\text{-}T_0$	Speed Measurement Filter Time Constant [s]	0.02 – 1s	0.02s	
$AFC\text{-}b_p$	Frequency influence (droop) bp	0.001 – 0,1 (0.1% – 10%)	0.05 (5%)	
AFC-SW	Frequency influence software switch	0 – 1	1	0 = PID 1 = bp
SC-SW	Synchronous condenser software switch	0 – 1	0	0 = Off 1 = On
SC_{sp}	Flow set point in Synchronous condenser mode	0 – 100%	0%	
Flow control mode parameters				

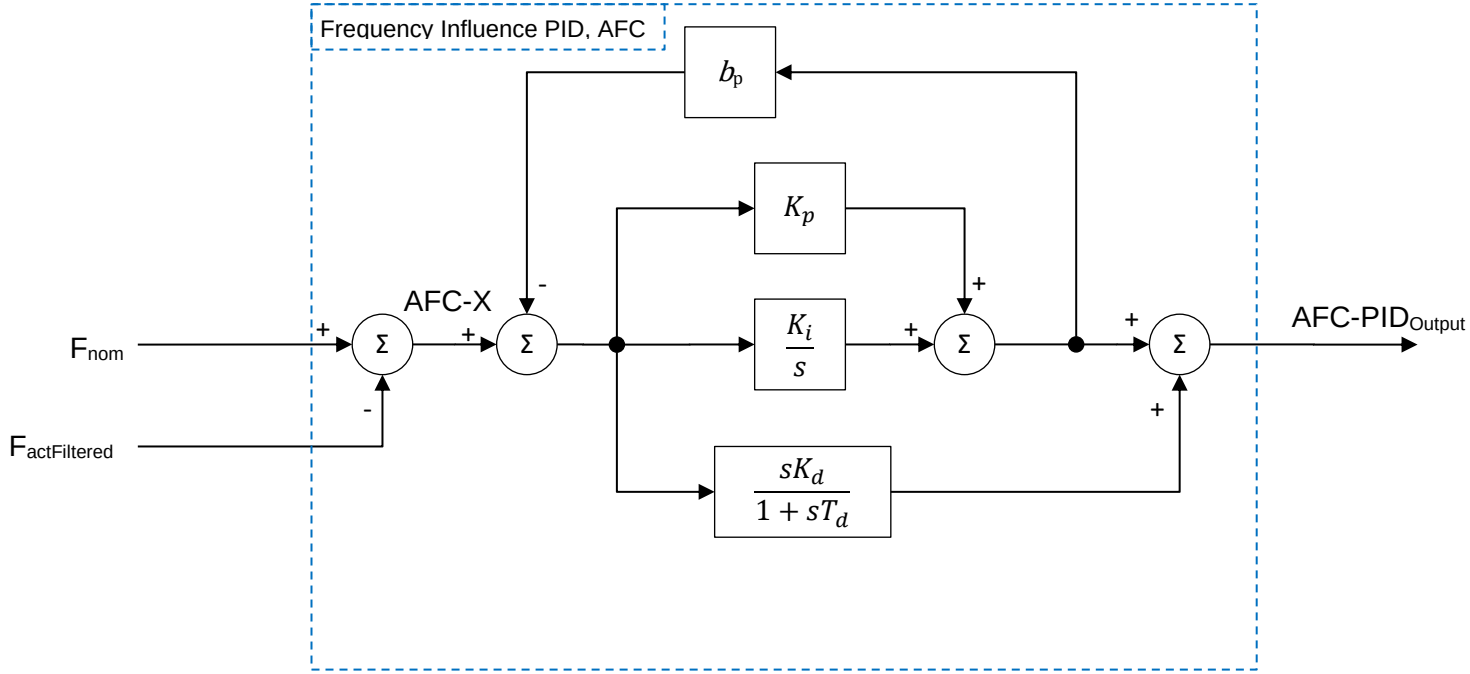
Transfer function for evaluation of the response to frequency disturbances for frequency influence with static droop

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{1}{AFC\text{-}b_p}$$

Transfer function for evaluation of the response to frequency disturbances for frequency influence with PID

$$\frac{\mathrm{FLOW_{sp}}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s} + \frac{s * AFC\text{-}K_d}{1 + s * AFC\text{-}T_d}}{1 + AFC\text{-}b_p \left(AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s}\right)}$$

7.4 Frequency Influence PID loop

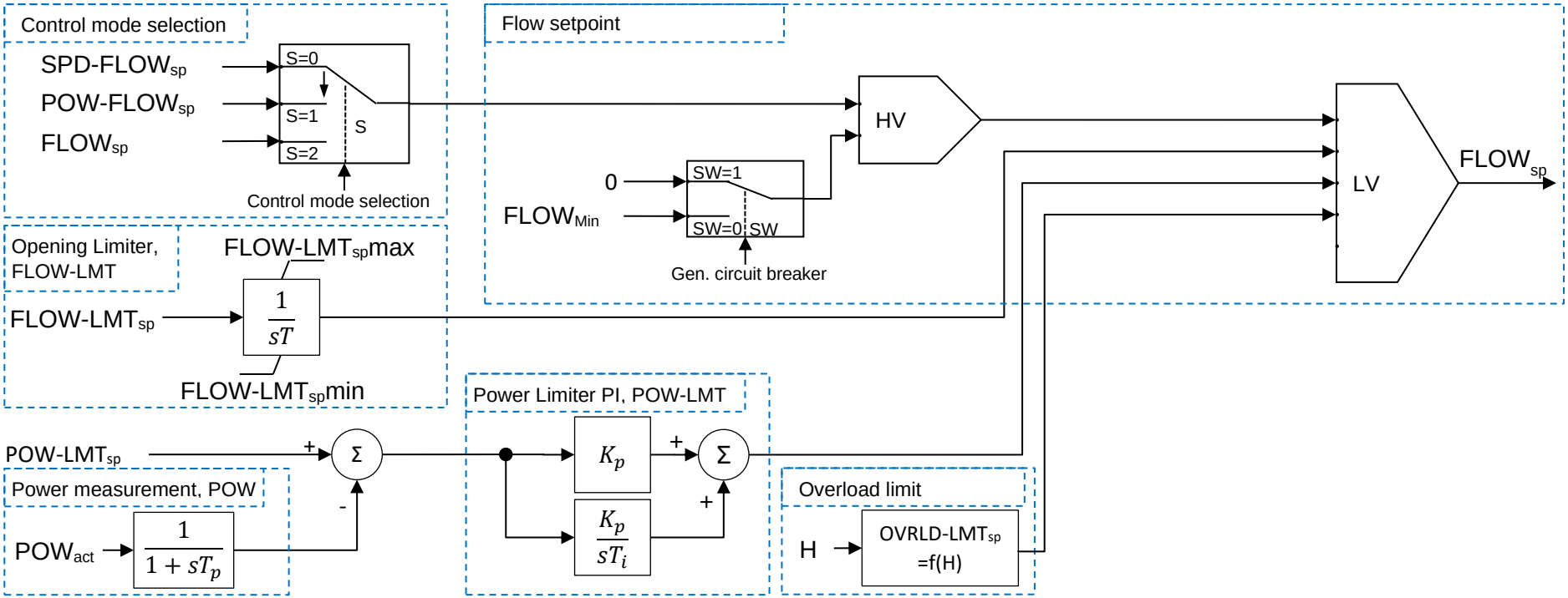


Parameters		Parameter Range	Typical	Comments
$AFC\text{-}bp$	Permanent droop	0 – 0.1 (0% - 10%)	0.05 (5%)	
$AFC\text{-}K_p$	Proportional gain	0.1 – 20	1.5	
$AFC\text{-}K_i$	Integral gain	0.01 – 100	0.5	
$AFC\text{-}K_d$	Derivative gain	0 – 10	0	
$AFC\text{-}T_d$	Time constant of the damping device [s]	0.1 – 10s	0s	
Frequency influence PID parameters				

Transfer function Frequency Influence PID:

$$\frac{AFC\text{-}PID_{Output}}{AFC\text{-}X} = \frac{AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s} + \frac{s * AFC\text{-}K_d}{1 + s * AFC\text{-}T_d}}{1 + AFC\text{-}b_p\left(AFC\text{-}K_p + \frac{AFC\text{-}K_i}{s}\right)}$$

7.5 Control signal selection



Parameters		Parameter Range	Typical	Comments
FLOW-LMT$_{sp}$min	Minimum allowed Flow Limiter set point	0 - 100%	0%	
FLOW-LMT$_{sp}$max	Maximum allowed Flow Limiter set point	0 - 100%	100%	
FLOW-LMT-T	Setpoint ramp time 0%-100% for Flow Limiter [s]	10 – 120s	60s	
POW-T$_p$	Power Measurement Filter Time Constant [s]	0.02 – 1s	0.2s	
FLOW$_{Min}$	Minimum flow with generator circuit breaker closed	0 – 10%	2% + O_{NL}	
POW-LMT-K$_p$	Proportional gain – Power Limiter	0.1 – 20	0.8	
POW-LMT-K$_i$	Integral gain – Power Limiter	0.01 – 100	0.15	
POW-LMT$_{sp}$	Maximum Power Limit	0 - 100%	100%	
OVRLD-LMT$_{sp}$	Corresponding overload flow limit to Head H	-		11 interpolated points
H	Actual head	-		
LV	Low value (Lowest value is transferred to output)	-		
HV	High value (Highest value is transferred to output)	-		
Control signal selection parameters				